SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Section	1
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Case Study	Member Management Module
System Name	KKK Online System
Group Name	Tech Hi-Five

Section A – System's Overview

1.1 Overall Description

The proposed version of the KKK Online System is to improve cooperative procedures such as member registration, loan application management, administrative tasks, approval tracking and financial reporting. By digitizing manual processes and enabling online registration, loan applications and approval tracking, it improves accessibility and efficiency. Board members look over, approve applications, manage data and policies. My module is admin approval, which is mostly focused on handling the approval process of the membership and loan application.

The module distribution of our project is shown as below:

Group Members' Name	**Matric No**	**Module**
Tan Yi Ya	A23CS0187	User Module
Chua Jia Lin	A23CS0069	Member Module
Teh Ru Qian	A23CS0191	Loan Module, Admin Reporting Module
Goe Jie Ying	A23CS0224	Admin Approval Module
Lam Yoke Yu	A23CS0233	Admin Updating Module

1.2 Project Weaknesses and Improvement

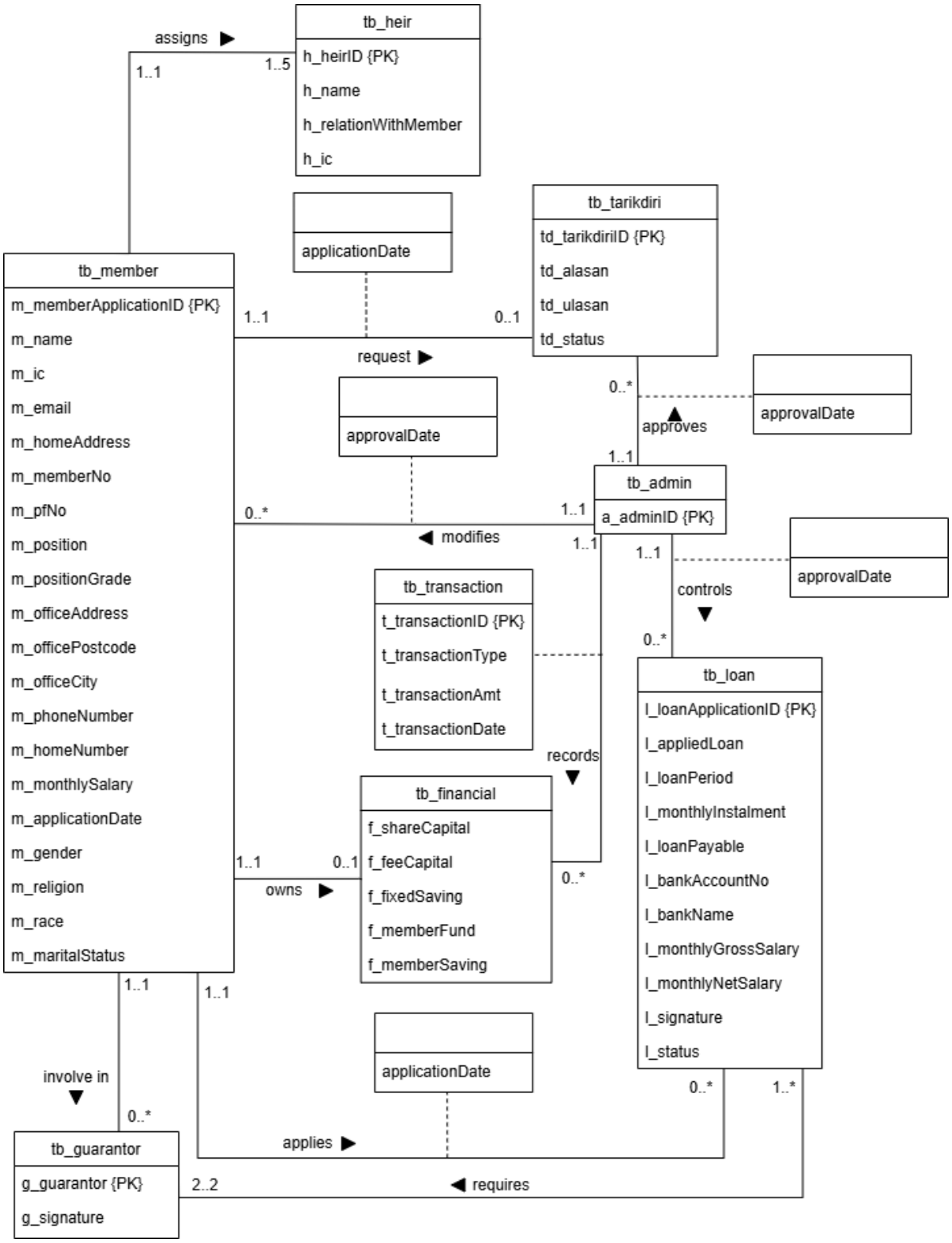
In the current implementation of the project, the functionality for admin to membership management is limited to approving the members and viewing the list of members after approving. However, for the approved member, they are added to the list, but admin cannot perform any further actions related to that member.

1.3 Proposed Module

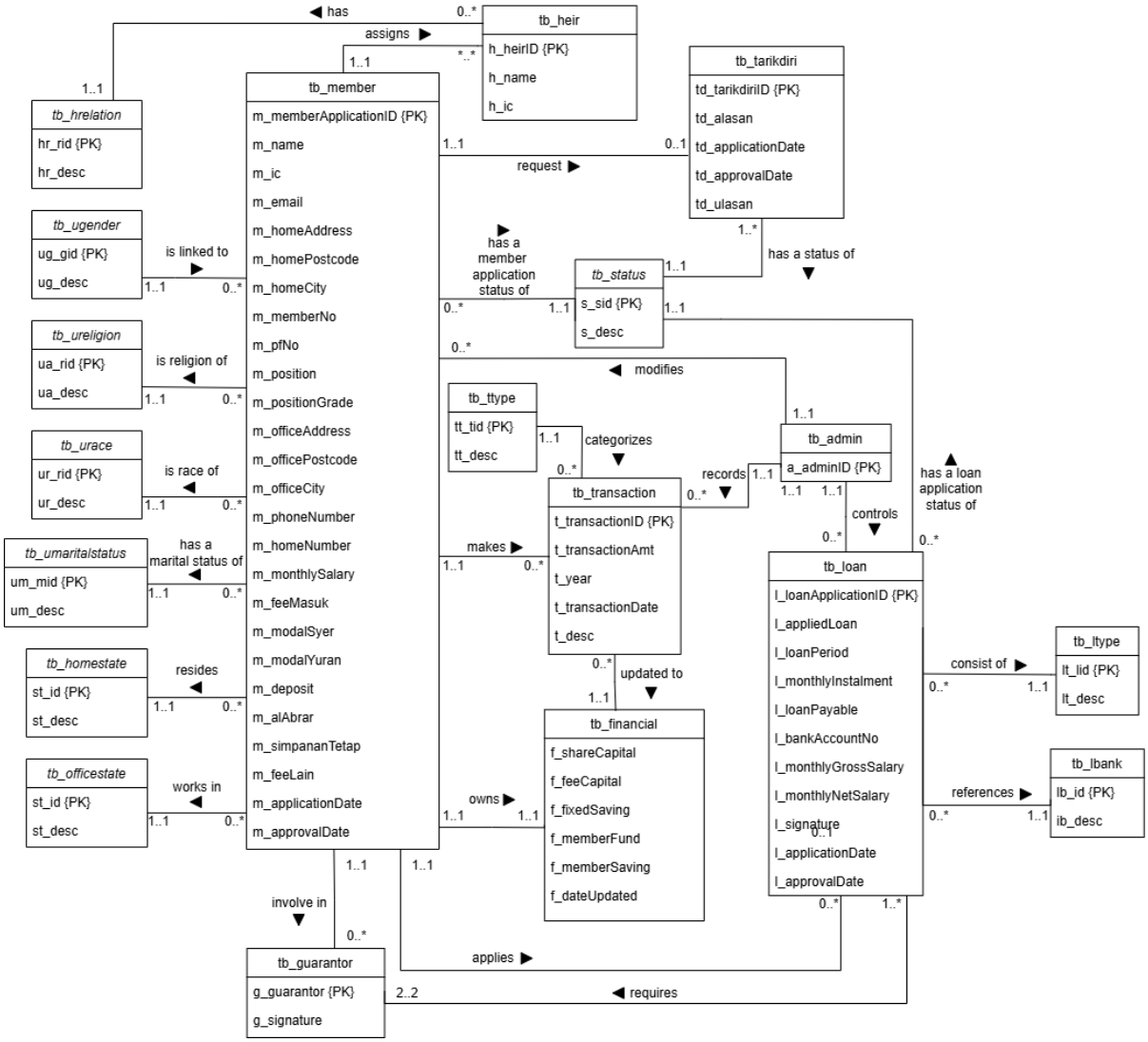
To enhance this functionality, a member management module should be developed. This new module allows admin to remove members who wish to quit the cooperative or who have retired from the list of members to ensure the list up-to-date and accurate.

Section B – DATABASE PLANNING AND DESIGN

2.1 Conceptual ERD



2.2 Logical ERD



2.2.1 Data Dictionary

2.2.1.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
tb_member	General term describing all user logged in as member using KKK Online System	KKK Member, Member Application	Each member has one gender, one religion, one race and one marital status only. Each member resides and works in one state. Each member can become guarantors for other members. Each member applies for many loans. Each member application will have one status only at any one time. Each member owns at most one financial record. Each member application is approved by

			one admin. Each member can make many transactions. Members are one kind of user for the system. Each member can assign many heirs. Each member may not or may request once only for stop being KKK member.
tb_ugender	General term describing the gender of a member	Gender	Each gender can be shared by many members.
tb_ureligion	General term describing the religion of a member	Religion	Each religion can be shared by many members.
tb_urace	General term describing the race of a member	Race	Each race can be shared by many members.
tb_umaritalstatus	General term describing the marital status of a member	Marital Status	Each marital status can be shared by many members.
tb_homestate	General term describing the home state of a member	State	There can be none or many members residing in the state.
tb_officestate	General term describing the office state of a member	State	There can be none or many members working in the state.
tb_admin	General term describing all user logged in as admin using KKK Online System	KKK Staff	Admin is one type of user. Admin can edit many banners. Each admin can generate many reports. Each admin alters many policies. Each admin controls many loan applications. Each admin approves many member applications. Each admin can record many transaction records.
tb_heir	General term describing all people declared as heir of a	-	Each heir is assigned to one member. Each heir for a single member will have a relationship with members only.

	member applicant		
tb_hrelation	General term describing the relationship between a member applicant and his heir	Relationship	Each heir relation with a member may be having by many members.
tb_transaction	General term describing all transaction made by member	-	Each transaction has one transaction type only. Each transaction record occurs in one month. Each transaction record will be recorded by a single admin only. Each transaction made by one member. Each transaction record will be updated to a member's financial record only.
tb_ttype	General term describing all types of transactions	Transaction type	Each transaction type belongs to many transactions.
tb_financial	General term describing all financial status of a member	-	Each financial record is owned by one member only. Each financial record can be updated by many transaction records.
tb_loan	General term describing all loans of the cooperative	Loan Application	Each loan is controlled by one admin. Each loan consists of one loan type. Each loan references a single bank only. Each loan application has a status at one time. Each loan application is applied by one member. Each loan application required two guarantors.
tb_ltype	General term describing all types of loans of the cooperative	-	Each loan type is being referenced by one or more loan applications.
tb_lbank	General term describing all banks	-	Each bank is being referenced by one or more loan applications.

tb_guarantor	General term describing all guarantors for a loan applicant	-	Two guarantors are required for each loan application. Each guarantor is involved by one member only.
tb_status	General term describing all status for the applications	-	Each membership and loan application has a single status at any one time.
tb_tarikdiri	General term describing all application to stop being a KKK member	Stop Being Member Application	Each stop being member application can be requested by a member only.

2.2.1.2 Identifying Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
tb_member	1..* 1..1 1..1 1..1 1..1 0..* 0..* 0..* 0..* 0..* 0..* 0..* 1..1	assigns involve in applies owns makes links is religion of is race of has a marital status of resides works in has a member application status of request	tb_heir tb_guarantor tb_loan tb_financial tb_transaction tb_ugender tb_ureligion tb_urace tb_umaritalstatus tb_homestate tb_officestate tb_status tb_tarikdiri	*..* 0..* 0..* 0..1 0..* 1..1 1..1 1..1 1..1 1..1 1..1 1..1 0..*
tb_admin	1..1 1..1 1..1	controls record modifies	tb_loan tb_financial tb_member	0..* 0..* 0..*
tb_heir	0..*	has	tb_hrelation	1..1
tb_loan	1..1 0..* 0..* 0..*	includes consists of references has a loan status of	tb_guarantor tb_ltype tb_lbank tb_status	2..2 1..1 1..1 1..1
tb_transaction	0..*	categorizes	tb_ttype	1..1

	0..* 0..*	updated to contains	tb_financial tb_month	1..1 1..1
tb_tarikdiri	1..*	has a status of	tb_status	1..1

2.2.1.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
tb_admin	a_adminID (PK)	int (11)	Uniquely identifies each admin	No	No	200
tb_member	memberApplicationID (PK)	int (11)	Uniquely identifies each member application	No	No	1001
	m_name	varchar(50)	User's full name	No	No	Ali bin Abu
	m_ic	varchar(14)	User's identification number	No	No	850101-08-1033
	m_homeAddress	varchar(255)	User's home address	No	No	1, Taman Bahagia, Jalan Bahagia,
	m_homePostcode	int (11)	User's home postcode	No	No	57000
	m_homeCity	varchar (50)	User's home city	No	No	Kuala Lumpur
	m_memberNo	int	Uniquely identifies each member	No	No	1
	m_pfNo	int	User's PF number	No	No	1001
	m_position	varchar(50)	User's position	No	No	Pengurus

	m_positionGrade	varchar(11)	User's position grade	No	No	G41
	m_officeAddress	varchar(255)	User's office address	No	No	25, Jalan Makmur
	m_officePostcode	int (11)	User's office postcode	No	No	57000
	m_officeCity	varchar (50)	User's office city	No	No	Kuala Lumpur
	m_phoneNumber	varchar(13)	User's phone number	No	No	0123456789
	m_homeNumber	varchar(13)	User's home number	Yes	No	051234567
	m_monthlySalary	double	User's monthly salary	No	No	3500.00
	m_feeMasuk	double	User's entry fee	No	No	45.00
	m_modalSyer	double	User's shares	No	No	300.00
	m_modalYuran	double	User's share fee	No	No	50.00
	m_deposit	double	User's deposit	No	No	50.00
	m_alAbrar	double	A kind of user's fund	No	No	5.00
	m_feeLain	doube	User's other fee	No	No	0.00
	m_applicationDate	datetime	The date when user submit his/her member application	No	No	2025-01-03 16:00:00
	m_approvalDate	timestamp	The date when the member application being approved by admin	Yes	No	2025-01-15 06:40:00
tb_ugender	ug_gid (PK)	int (11)	The unique id for gender	No	No	1

	ug_desc	varchar (10)	User's gender description	No	No	Lelaki
tb_urelig ion	ua_rid (PK)	int (11)	The unique id for religion	No	No	1
	ua_desc	varchar (10)	User's religion description	No	No	Islam
tb_urace	ur_rid (PK)	int (11)	The unique id for race	No	No	1
	ur_desc	varchar (10)	User gender's description	No	No	Lelaki
tb_umari talstatus	um_mid (PK)	int (11)	The unique id for marital status	No	No	1
	um_desc	varchar (10)	User's marital status description	No	No	Bujang
tb_home state	st_id (PK)	int (11)	The unique id for user's home state	No	No	1
	st_desc	varchar (15)	User's home state description	No	No	Johor
tb_office state	st_id (PK)	int (11)	The unique id for user's office state	No	No	1
	st_desc	varchar (15)	User's office state description	No	No	Melaka
tb_heir	h_heirID (PK)	int (11)	Uniquely identifies each heir	No	No	1
	h_name	varchar(50)	Heir's full name	No	No	Tan Mei Ling
	h_ic	varchar(14)	Heir's identification number	No	No	010101-01-0101
tb_hrelat ion	hr_rid (PK)	int (11)	Uniquely identifies each heir relationship with member	No	No	1
	hr_desc	varchar (20)	The description of heir relationship with member	No	No	Suami Isteri

tb_transaction	t_transactionID (PK)	int	Uniquely identifies each transaction	No	No	1
	t_transactionAmt	double	Transaction amount	No	No	100
	t_year	int (11)	The year when the transaction made	No	No	2024
	t_transactionDate	timestamp	Transaction date	No	No	2024-12-20 17:30:00
	t_desc	varchar (255)	The objective of the transaction made	NO		Potongan Gaji Bayaran Balik 8
tb_ttype	tt_lid (PK)	int	Uniquely identifies each transaction type	No	No	1
	tt_desc	varchar (30)	The description of each transaction type	No	No	Al-Qadrul Hassan
tb_financial	f_shareCapital	double	Member’s share capital	No	No	10
	f_feeCapital	double	Member’s fee capital	No	No	100
	f_fixedSaving	double	Member’s fixed saving	No	No	200
	f_memberFund	double	Member’s fund	No	No	50
	f_memberSaving	double	Member’s saving	No	No	200
	f_dateUpdated	double	The date when the financial record is updated	No	No	2025-01-21 15:36:25
tb_loan	l_loanApplicationID (PK)	int	Uniquely identifies each loan application	No	No	1

	l_appliedLoan	double	Amount of loan applied	No	No	10000
	l_loanPeriod	int	Loan period	No	No	4
	l_monthlyInstalment	double	Monthly instalment for loan	No	No	90.33
	l_loanPayable	double	The total loan that need to be paid by the member	No	No	1897.01
	l_bankAccountNo	int	Loan borrower's bank account number	No	No	214748364
	l_monthlyGrossSalary	double	Loan borrower's monthly gross salary	No	No	10000
	l_monthlyNetSalary	double	Loan borrower's monthly net salary	No	No	8000
	l_signature	varchar(50)	Loan borrower's signature	No	No	./uploads/8d969f4e9e3586a1226cc389a6a29b1b.jpeg
	l_applicationDate	datetime	The date where loan application is made by a member	No	No	2025-01-05 12:51:36
	l_approvalDate	timestamp	The date where loan application is approved by admin	Yes	No	2025-01-11 23:40:25
tb_ltype	lt_lid (PK)	int	Uniquely identifies each retrieval record	No	No	1
	lt_desc	varchar (30)	The description of each loan type	No	No	Al-Qadrul Hassan

tb_lbank	lb_lid (PK)	int	Uniquely identifies each bank	No	No	1
	lb_desc	varchar (30)	The description of each bank	No	No	BSN
tb_guarantor	g_guarantorID (PK)	int (11)	Uniquely identifies each guarantor	No	No	1
	g_signature	varchar(50)	Guarantor's signature	No	No	./uploads/678a24a16eee9.jpeg
tb_status	s_sid (PK)	int (11)	Uniquely identifies each status	No	No	2
	s_desc	varchar (20)	The description of each status	No	No	Ditolak
tb_tarikdiri	td_tarikdiriID (PK)	int (11)	Uniquely indentifies each stop being member application	No	No	1
	td_alasan	varchar (255)	The reason of application for stop being member that filled by the member.	No	No	Not longer a employee for KADA
	td_applicationDate	datetime	The date where application is made by a member	No	No	2025-01-05 12:51:36
	td_approvalDate	timestamp	The date where the application is approved by admin	Yes	No	2025-01-11 23:40:25
	td_ulasan	Varchar (255)	The comment filled by the KKK staff if they wish to reject the request	Yes	No	Terdapat baki yang belum dibayar

	td_status	int (11)	The status of the stop being member application	No	No	1

2.3 Relational Database Schemes

tb_admin (a_adminID)
PK: a_adminID

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
FK: f_memberNo references tb_member (m_memberNo)

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)
PK: g_guarantorID
FK: g_loanApplicationID references tb_loan (l_loanApplicationID)
FK: g_memberNo references tb_member (m_memberNo)

tb_heir (h_heirID, h_memberApplicationID, h_name, h_relationWithMember, h_ic)
PK: h_heirID
FK: h_memberApplicationID references tb_member (m_memberApplicationID)
FK: h_relationWithMember references tb_hrelation (hr_rid)

tb_hrelation (hr_rid, hr_desc)
PK: hr_rid

tb_loan (l_loanApplicationID, l_memberNo, l_loanType, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, l_bankName, l_monthlyNetSalary, l_signature, l_file, l_status, l_applicationDate, l_approvalDatae, l_adminID)
PK: l_loanApplicationID
FK: l_loanType references tb_ltype (lt_lid)
FK: l_status references tb_status (s_sid)
FK: l_memberNo references tb_member (m_memberNo)
FK: l_bankName references tb_lbank (lb_id)
FK: l_adminID references tb_admin (a_adminID)

tb_lbank (lb_id, lb_desc)
PK: lb_id

tb_ltype (lt_lid, lt_desc)
PK: lt_lid

tb_member (m_memberApplicationID, m_name, m_ ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID
FK: m_gender references tb_ugender (ug_gid)
FK: m_religion references tb_ureligion (ua_rid)
FK: m_race references tb_urace (ur_rid)
FK: m_maritalStatus references tb_umaritalstatus (um_mid)
FK: m_status references tb_status (s_sid)
FK: m_homeState references tb_homestate (st_id)
FK: m_officeState references tb_officestatus (st_id)
FK: m_adminID references tb_admin (a_adminID)

tb_ugender (ug_gid, ug_desc)
PK: ug_gid

tb_ureligion (ua_rid, ua_desc)
PK: ua_rid

tb_urace (ur_rid, ur_desc)
PK: ur_rid

tb_umaritalstatus (um_mid, um_desc)
PK: um_mid

tb_status (s_sid, s_desc)
PK: s_sid

tb_homestate (st_id, st_desc)
PK: st_id

tb_offisestate (st_id, st_desc)
PK: st_id

tb_transaction (t_transactionID, t_transactionType, t_transactionAmt, t_month, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)
PK: t_transactionID
FK: t_transactionType references tb_ttype (tt_id)
FK: t_month references tb_rmonth (rm_id)
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

tb_ttype (tt_id, tt_desc)
PK: tt_id

tb_tarikdiri (td_tarikdiriID, td_alasan, td_applicationDate, td_approvalDate, td_ulasan, td_status)
PK: td_tarikdiriID
FK: td_status references tb_status (s_sid)

2.4 Normalization

1NF
tb_admin (adminID)
PK: adminID

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)
PK: g_guarantorID
FK: g_loanApplicationID references tb_loan (l_loanApplicationID)
FK: g_memberNo references tb_member (m_memberNo)

tb_loan (l_loanApplicationID, l_memberNo, lt_lid, lt_desc, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, lb_id, lb_desc, l_monthlyNetSalary, l_signature, l_file, s_sid, s_desc, l_applicationDate, l_approvalDate, l_adminID)
PK: l_loanApplicationID
FK: l_memberNo references tb_member (m_memberNo)
FK: l_adminID references tb_admin (a_adminID)

tb_tarikdiri (td_tarikdiriID, td_alasan, td_applicationDate, td_approvalDate, td_ulasan, td_status)
PK: td_tarikdiriID
FK: td_status references tb_status (s_sid)

tb_member (m_memberApplicationID, m_name, m_ ic, m_email, ug_gid, ug_desc, ua_rid, ua_desc, ur_rid, ur_desc, um_mid, um_desc, m_homeAddress, m_homePostcode, m_homeCity, st_id, st_desc, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, st_id, st_desc, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, s_sid, s_desc, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (a_adminID)

tb_heir (h_heirID, h_memberApplicationID, h_name, hr_rid, hr_desc, h_ic)
PK: h_heirID

tb_transaction (t_transactionID, tt_id, tt_desc, t_transactionAmt, rm_id, rm_desc, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)
PK: t_transactionID
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
PK: f_memberNo
FK: f_memberNo references tb_member (m_memberNo)

2NF

tb_admin (adminID)
PK: adminID

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)
PK: g_guarantorID
FK: g_loanApplicationID references tb_loan (l_loanApplicationID)
FK: g_memberNo references tb_member (m_memberNo)

tb_loan (l_loanApplicationID, l_memberNo, lt_lid, lt_desc, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, lb_id, lb_desc, l_monthlyNetSalary, l_signature, l_file, s_sid, s_desc, l_applicationDate, l_approvalDate, l_adminID)
PK: l_loanApplicationID
FK: l_memberNo references tb_member (m_memberNo)
FK: l_adminID references tb_admin (a_adminID)

tb_tarikdiri (td_tarikdiriID, td_alasan, td_applicationDate, td_approvalDate, td_ulasan, td_status)
PK: td_tarikdiriID
FK: td_status references tb_status (s_sid)

tb_member (m_memberApplicationID, m_name, m_ ic, m_email, ug_gid, ug_desc, ua_rid, ua_desc, ur_rid, ur_desc, um_mid, um_desc, m_homeAddress, m_homePostcode, m_homeCity, st_id, st_desc, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, st_id, st_desc, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, s_sid, s_desc, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (a_adminID)

tb_heir (h_heirID, h_memberApplicationID, h_name, hr_rid, hr_desc, h_ic)
PK: h_heirID

tb_transaction (t_transactionID, tt_id, tt_desc, t_transactionAmt, rm_id, rm_desc, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)
PK: t_transactionID
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
PK: f_memberNo
FK: f_memberNo references tb_member (m_memberNo)

3NF

tb_admin (adminID)
PK: adminID

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)
PK: g_guarantorID
FK: g_loanApplicationID references tb_loan (l_loanApplicationID)
FK: g_memberNo references tb_member (m_memberNo)

tb_loan (l_loanApplicationID, l_memberNo, l_loanType, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, l_bankName, l_monthlyNetSalary, l_signature, l_file, l_status, l_applicationDate, l_approvalDate, l_adminID)
PK: l_loanApplicationID
FK: l_memberNo references tb_member (m_memberNo)
FK: l_adminID references tb_admin (adminID)
FK: l_loanType references tb_ltype (lt_lid)
FK: l_bankName references tb_lbank (lb_id)
FK: l_status references tb_status (s_sid)

tb_tarikdiri (td_tarikdiriID, td_alasan, td_applicationDate, td_approvalDate, td_ulasan, td_status)
PK: td_tarikdiriID
FK: td_status references tb_status (s_sid)

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (adminID)
FK: m_gender references tb_ugender (ug_gid)
FK: m_religion references tb_urace (ua_rid)
FK: m_race references tb_ureligion (ur_rid)
FK: m_maritalStatus references tb_umaritalStatus (um_mid)
FK: m_homeState references tb_state (st_id)
FK: m_officeState references tb_state (st_id)
FK: m_status references tb_status (s_sid)

tb_heir (h_heirID, h_memberApplicationID, h_name, h_relationWithMember, h_ic)
PK: h_heirID
FK: h_memberApplicationID references tb_member (m_memberApplicationID)
FK: h_relationWithMember references tb_hrelation (hr_rid)

tb_transaction (t_transactionID, t_transactionType, t_transactionAmt, t_month, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)
PK: t_transactionID
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

FK: t_transactionType references tb_ttype (tt_id)
FK: t_month references tb_month (rm_id)
tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
PK: f_memberNo
FK: f_memberNo references tb_member (m_memberNo)

tb_ltype (lt_lid, lt_desc)
PK: lt_lid

tb_lbank (lb_id, lb_desc)
PK: lb_id

tb_status (s_sid, s_desc)
PK: s_sid

tb_ugender (ug_gid, ug_desc)
PK: ug_gid

tb_ureligion (ua_rid, ua_desc)
PK: ua_rid

tb_urace (ur_rid, ur_desc)
PK: ur_rid

tb_umaritalStatus (um_mid, um_desc)
PK: um_mid

tb_state (st_id, st_desc)
PK: st_id

tb_hrelation (hr_rid, hr_desc)
PK: hr_rid

tb_ttype (tt_id, tt_desc)
PK: tt_id

tb_month (rm_id, rm_desc)
PK: rm_id